Infosys Springboard Virtual Internship 7.0

Project Name: Supply Chain Visibility & Optimization

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1. Introduction

- Supply chain management:

Supply Chain Management involves planning, sourcing, manufacturing, transportation, warehousing, and delivering products efficiently from suppliers to customers. In today's competitive business environment, organizations require complete visibility across their supply chains to make informed decisions and improve operational performance.

- Supply chain Visibility:

Supply Chain Visibility refers to the ability to monitor inventory, suppliers, warehouses, transportation activities, and customer deliveries in real time. Improved visibility enables organizations to reduce costs, identify bottlenecks, optimize inventory levels, and enhance customer satisfaction.

2. Problem Statement

Many organizations face difficulties in managing inventory, supplier coordination, warehouse operations, manufacturing processes, and transportation networks due to limited visibility across the supply chain.

These challenges often result in:

- Stock shortages
- Delayed deliveries
- Increased operational costs
- Poor inventory planning
- Reduced customer satisfaction

3.Objectives

The primary objectives of this project are:

- **Analyze the Data**

To understand inventory availability, supplier performance, manufacturing activities, logistics operations, and customer demand patterns.

- **Preprocess the Dataset**

To handle missing values, remove duplicate records, correct data types, and prepare a clean dataset for analysis.

- **Build Interactive Dashboards**

To develop dashboards in Microsoft Power BI that effectively visualize key performance indicators (KPIs) and supply chain metrics.

- **Generate Actionable Insights**

To provide business insights that support inventory optimization, supplier evaluation, logistics management, and strategic decision-making.

4. Dataset Description

The dataset used in this project contains:

- 1,540 records
- 24 key attributes

The data covers multiple areas of supply chain operations, including:

i. **Product Information**
   - Product Type
   - SKU
   - Price
   - Availability
   - Products Sold
   - Revenue Generated
   - Stock Levels
   - Order Quantities

ii. **Logistics Data**
   - Shipping Time
   - Shipping Carrier
   - Shipping Cost
   - Transportation Mode
   - Route Information

iii. **Manufacturing and Supplier Data**
   - Supplier Name
   - Production Volume
   - Manufacturing Cost
   - Lead Time
   - Inspection Results
   - Defect Rate

5. Data Import and Understanding:

In this stage, the raw supply chain dataset was imported into Python using the Pandas library. The dataset structure, dimensions, column names, and data types were examined to gain a clear understanding of the data. Numerical and categorical features were reviewed, and data quality checks were performed to ensure the dataset was suitable for preprocessing and further analysis.

Example Python functions used:

Function	**Description**
df.head()	Displays the first 5 rows of the dataset.
df.shape	Shows the number of rows and columns in the dataset
df.columns	Lists all column names present in the dataset.
df.info()	Displays data types, non-null counts, and dataset information
df.duplicates()	list all duplicate records in the dataset

6. Data Preprocessing

- **Data Import**

The dataset was imported into Python using the Pandas library to perform preprocessing and analysis.

- **Data Inspection**

The dataset structure, dimensions, column names, and data types were examined using functions such as head(), shape(), info(), and describe().

- **Missing Value Analysis**

The dataset was checked for null or missing values to identify incomplete records that could affect analysis.

- **Missing Values filling**

Missing values were handled using appropriate techniques such as replacing categorical values with meaningful labels and numerical values with suitable statistical measures.

- **Duplicate Record Detection**

The dataset was examined for duplicate entries to prevent inaccurate calculations and reporting.

- **Duplicate Removal**

Identified duplicate records were removed to improve data quality and consistency.

- **Data Type Validation**

Each column's data type was verified to ensure that numerical, categorical, and date fields were stored correctly.

- **Data Type Conversion**

Columns were converted into appropriate formats such as numeric, date, or text data types wherever required.

- **Data Cleaning and standardization**

Unwanted spaces, inconsistent values, and formatting issues were removed to standardize the dataset..

- **Power BI Integration**

The cleaned dataset was loaded into Microsoft Power BI for visualization, reporting, and insight generation.

7. Tools and Technologies Used

Python

Python was used for:

- Data import
- Data cleaning
- Missing value treatment
- Duplicate removal
- Data transformation
- Dataset preparation

Microsoft Power BI

Power BI was used for:

- Dashboard development
- KPI visualization
- Supply chain performance monitoring
- Business insight generation

8.Learning Outcomes :

Through the data preprocessing phase of this project, I gained practical experience in preparing raw data for analysis and visualization. I learned how to import datasets into Python, examine dataset structures, and understand the importance of data quality in analytics projects.

Key learnings include:

- Learned how to import and explore datasets using Python.
- Identified and handled missing values effectively.
- Detected and removed duplicate records.
- Performed data type validation and conversion.

- Cleaned inconsistent and unnecessary data.
- Standardized column names and data formats.
- Used Pandas and NumPy for data preprocessing.

9. Expected Outcomes

The project delivers the following benefits:

- Improved visibility across supply chain operations
- Better inventory management
- Enhanced supplier performance evaluation
- Faster decision-making through dashboards
- Reduced operational inefficiencies
- Increased customer satisfaction
- Data-driven business optimization

10. Conclusion

Supply Chain Visibility & Optimization demonstrates how data analytics can improve supply chain performance by transforming raw operational data into meaningful insights. Through systematic data preprocessing in Python and visualization in Microsoft Power BI, the project enables organizations to monitor key supply chain activities, identify inefficiencies, and make informed decisions.

The developed solution supports inventory optimization, supplier assessment, logistics monitoring, and strategic planning, ultimately contributing to greater operational efficiency and customer satisfaction.